ModelIt!

Teacher's Guide

Can Nuclear Power Actually Save the Planet?

Modeling Nuclear Fission, Energy Output, and Environmental Trade-offs

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-PS1-8, HS-ESS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS1-8, HS-ESS3-2

Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay. Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to illustrate the energy released during fission and evaluate the environmental trade-offs of nuclear power generation.
Disciplinary Core Idea	**PS1.C Nuclear Processes / ESS3.A Natural Resources** — Nuclear fission releases energy when heavy nuclei split, and evaluating nuclear power requires analyzing costs and benefits of energy resource management.
Crosscutting Concept	**Energy and Matter** — Students track energy transformations from nuclear binding energy to thermal energy to electrical energy, and trace matter flows from uranium fuel to fission products and waste.

CAST Testing Alignment

MULTIPLE CHOICE:

A nuclear power plant produces 8 billion kWh of electricity per year with near-zero CO_2 emissions. If this plant were replaced by a natural gas plant, approximately how much additional CO_2 would be released annually?

CONSTRUCTED RESPONSE:

Using your model, evaluate whether nuclear power should be expanded as a strategy to combat climate change. Discuss both the carbon emission benefits and the radioactive waste challenges, using model evidence to support your argument.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Nuclear energy occupies a unique position in the climate debate: it is the most energy-dense, lowest-carbon reliable power source available, yet it produces waste that remains dangerous for longer than human civilization has existed. Understanding the science of fission — and the genuine trade-offs it presents — is essential for making informed decisions about our energy future.

The Physics of Fission

When a neutron strikes a uranium-235 nucleus, the atom splits into two lighter elements (fission products), releases 2-3 additional neutrons, and converts a tiny amount of mass into enormous energy via Einstein's $E=mc^2$. A single fission event releases about 200 MeV — roughly 50 million times more energy than burning one molecule of methane. This is why nuclear fuel is so extraordinarily energy-dense.

Chain Reactions and Control

The neutrons released by one fission event can trigger additional fissions, creating a chain reaction. In a reactor, this chain reaction is carefully controlled using neutron-absorbing control rods — pushed in to slow the reaction, pulled out to speed it up. In a bomb, the chain reaction is uncontrolled. The key engineering challenge is maintaining criticality — exactly one neutron from each fission causing exactly one more fission.

The Waste Problem

Spent nuclear fuel contains fission products and transuranic elements that emit dangerous radiation. Some isotopes (like cesium-137) have half-lives of 30 years; others (like plutonium-239) have half-lives of 24,000 years. The US has accumulated about 83,000 metric tons of spent fuel with no permanent repository. Finland's Onkalo facility, designed to store waste for 100,000 years, is the world's first deep geological repository.

Nuclear vs. Fossil Fuels: The Numbers

Over its lifecycle, nuclear power emits about 12 grams of CO_2 per kWh — comparable to wind (11g) and far below solar (44g), natural gas (490g), and coal (820g). The world's 440 nuclear reactors prevent about 2 billion tons of CO_2 emissions annually. However, nuclear plants are expensive to build ($10-20 billion each), take 10-15 years to construct, and face significant public opposition due to accidents at Chernobyl (1986) and Fukushima (2011).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify uranium fuel supply, fission rate, energy output, radioactive waste, and carbon emissions avoided as the key components of the nuclear energy system.

E **Establish Relationships**
Students determine that uranium fuel supply enables fission, fission rate drives both energy output and waste production, and energy output directly determines how much carbon emission is avoided compared to fossil fuels.

V **Visualize & Model**
Students build a computational model showing the relationships between fuel input, fission rate, useful energy output, waste generation, and climate impact.

E **Evaluate Outcomes**
Students run full power, reduced power, and fuel depletion scenarios to quantify the trade-offs between energy production, waste generation, and carbon avoidance.

R **Revise & Extend**
Students add cooling water demand, fuel enrichment level, or public risk perception to model more complex real-world nuclear energy dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What if I told you there's an energy source that produces almost zero carbon, fits a year's fuel in a pickup truck, but creates waste that's dangerous for 100,000 years? Would you use it?"*

Watch For

Show the scale comparison: a uranium pellet vs. a ton of coal producing the same energy. Let students react.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the most controversial clean energy source on Earth — and you'll decide if it should be expanded."*

Watch For

Have students read objectives. Pre-teach 'fission' and 'half-life' with animations.

Slide 3: Big Question

3 min

Say: *"Nuclear produces almost zero carbon. So why isn't every country building reactors as fast as possible?"*

Watch For

Quick poll: Who thinks nuclear should be expanded? Who thinks it should be phased out? Record the split for later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the complete nuclear system — energy in, energy out, waste produced, carbon avoided."*

Watch For

Preview LEVER steps. Emphasize that this is about TRADE-OFFS, not right/wrong answers.

Slide 5: Activity 1: Components

8 min

Say: *"What goes INTO a nuclear plant and what comes OUT? Sort the inputs from the outputs."*

Watch For

Guide sorting. Uranium supply is the only true external input; fission rate is controlled but internal.

Slide 6: Activity 2: Connections

8 min

Say: *"More fission means more energy — but it also means more waste. Can you have one without the other?"*

Watch For

Help students map the dual outputs. Use green arrows for benefits, red for costs.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run a nuclear plant for a year and see exactly what we get — energy AND waste."*

Watch For

Students predict output ratios. Run full power, reduced, and depletion scenarios. Compare trade-offs.

Slide 8: Discoveries

5 min

Say: *"One uranium pellet equals one ton of coal. But that pellet also creates waste that outlasts the pyramids. How do we weigh that?"*

Watch For

Revisit the opening poll. Has anyone changed their mind? Discuss what evidence shifted perspectives.

Slide 9: STEM Challenge

12 min

Say: *"You're advising the President on the 2050 energy mix. 350 million people need power. Climate targets must be met. What's your plan?"*

Watch For

Groups design energy portfolios balancing nuclear, renewables, and fossil fuels. Present and defend choices.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Uranium Fuel Supply
- **INTERNAL:** Fission Rate, Energy Output, Radioactive Waste, Carbon Emissions Avoided

Relationships

- **Uranium Fuel Supply to Fission Rate:** POSITIVE (+) — More available uranium fuel allows for sustained or increased fission rates. As fuel is consumed, fewer fissile atoms remain, eventually requiring fuel replacement to maintain output.
- **Fission Rate to Energy Output:** POSITIVE (+) — Higher fission rates mean more atoms splitting per second, releasing more thermal energy to generate steam and drive turbines. Doubling the fission rate approximately doubles the electrical output.
- **Fission Rate to Radioactive Waste:** POSITIVE (+) — Higher fission rates produce more fission products and transuranic elements per unit time. Energy and waste are inseparable outputs of the fission process — you cannot have one without the other.

Key Discoveries

- Nuclear fission releases roughly 2 million times more energy per kilogram of fuel than burning coal — making it extraordinarily energy-dense and low-carbon
- Radioactive waste is produced in relatively small volumes (about 2,000 metric tons per year for the entire US fleet) but remains hazardous for extremely long periods
- The carbon emissions avoided by nuclear power are enormous — replacing a single nuclear plant with natural gas would add millions of tons of CO2 per year
- The nuclear debate is fundamentally a trade-off between a certain, manageable risk (waste storage) and a catastrophic, diffuse risk (climate change from fossil fuels)

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Nuclear power plants can explode like atomic bombs"	A nuclear reactor cannot produce a nuclear explosion. Reactor fuel is enriched to 3-5% uranium-235; a nuclear weapon requires 90%+ enrichment. The worst-case reactor accident is a meltdown — where fuel overheats and damages the containment — not a nuclear detonation. Chernobyl's explosion was a steam explosion from overheated water, not a nuclear blast.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Nuclear waste glows green and will poison the Earth forever"	Spent nuclear fuel doesn't glow green — that's a Hollywood invention. It's a solid ceramic material stored in steel and concrete casks. While it is genuinely dangerous due to radiation, the total volume is small (all US nuclear waste from 60 years fits on one football field, 10 yards deep). The longest-lived isotopes are actually the LEAST radioactive — the most dangerous waste decays within a few hundred years.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Renewable energy can completely replace nuclear without trade-offs"	Wind and solar are excellent but intermittent — the sun doesn't always shine and the wind doesn't always blow. Without massive battery storage (which doesn't yet exist at grid scale), removing nuclear baseload power means burning fossil fuels during renewable gaps. Germany's nuclear phase-out led to increased coal burning and higher emissions. The model shows that nuclear provides the firm, 24/7 clean power that renewables alone cannot yet deliver.